NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Urban Informatics

Assignment - Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **14** **Total marks: (8 X 1) + (6 X 2) = 20 marks**

MCQ/MSQ Question

QUESTION 1: [1 mark]

Which of the following machine learning models is inspired by the structure and functioning of the human brain?

a. Artificial Neural Network (ANN)
b. Decision Tree
c. K-Nearest Neighbours
d. Support Vector Machine

Correct Answer: a. Artificial Neural Network (ANN)

Detailed explanation: Artificial Neural Network (ANN) is a machine learning model inspired by the structure and functioning of the human brain.
Refer to lecture 46, slide 3.

QUESTION 2: [1 mark]

In an artificial neural network, the number of neurons in the hidden layer mainly depends on which of the following?

a. The size of the input dataset
b. The number of decision boundaries required to separate different classes
c. The number of output classes only
d. The number of training iterations

Correct Answer: b. The number of decision boundaries required to separate different classes

Detailed explanation: Number of neurons in the hidden layer depends on the number of decision boundaries we need to draw between the different classes/categories that need to be predicted.
Refer to lecture 46, slide 4.

QUESTION 3: [1 mark]

_______________ is used in modern deep learning architecture such as transformers.

a. ELU (Exponential Linear Unit)
b. Softmax
c. GELU (Gaussian Error Linear Unit)
d. ReLU (Rectified Linear Unit)

Correct Answer: c. GELU (Gaussian Error Linear Unit)

Detailed explanation: GELU is used in modern deep learning architecture such as transformers. Softmax is used for multi-class classification problems. ELU is used where smoother learning and faster convergence are needed. ReLU is used for general deep learning architectures.
Refer to lecture 46, slide 6.

QUESTION 4: [1 mark]

Backpropagation in a neural network is primarily used for which of the following?

a. Increasing the number of hidden layers
b. Initializing the weights of the network
c. Computing the partial derivatives of the loss function with respect to weights and biases
d. Reducing the size of the training dataset

Correct Answer: c. Computing the partial derivatives of the loss function with respect to weights and biases
Detailed explanation: Backpropagation is a systematic procedure for computing the partial derivatives of the loss function with respect to every weight and bias in the network.
Refer to lecture 46, slide 9.

0QUESTION 5: [1 mark]

Which of the following best describes the purpose of SHAP in machine learning models?

a. It improves the training speed of neural networks
b. It replaces the loss function in deep learning models
c. It increases the number of hidden layers in a neural network
d. It explains the contribution of each feature to a model's prediction using Shapley values

Correct Answer: d. It explains the contribution of each feature to a model's prediction using Shapley values

Detailed explanation: SHAP is an explainable AI method based on Shapley values from cooperative game theory. It explains how much each feature contributes to a model's prediction. ANNs are black-box models: SHAP makes them interpretable.
Refer to lecture 46, slide 10.

QUESTION 6: [1 mark]

Why are loss surfaces in deep neural networks highly complex?

a. Because neural networks use only linear functions
b. Because neural networks always converge to a single global minimum
c. Due to nonlinear activation functions, multiple layers, and a large number of parameters
d. Due to the absence of optimization algorithms

Correct Answer: c. Due to nonlinear activation functions, multiple layers, and a large number of parameters

Detailed Solution: Loss surfaces are very complex due to nonlinear activation functions, multiple layers and millions of parameters. It contains local minima, saddle points, flat plateaus and sharp minima which needs to be overcome.
Refer to lecture 47, slide 5.

QUESTION 7: [2 marks]

Which of the following are major challenges in deep learning optimization?

P. Vanishing gradients
Q. Exploding gradients
R. Weight initialisation
S. Batch normalisation

a. P and Q
b. P and R
c. Q and S
d. Q and R

Correct Answer: a. P and Q

Detailed Solution: Refer to lecture 47, slide 8 and 9.

QUESTION 8: [2 marks]

Which of the following correctly describes the primary use of CNNs and RNNs?

a. CNNs are used for sequential data, while RNNs are used for spatial data
b. Both CNNs and RNNs are used only for image classification
c. CNNs and RNNs are used only for unsupervised learning
d. CNNs are used for spatial data, while RNNs are used for sequential/temporal data

Correct Answer: d. CNNs are used for spatial data, while RNNs are used for sequential/temporal data

Detailed Solution: Convolutional Neural Networks (CNNs) are used for spatial data (CNNs learn "where" patterns occur) while Recurrent Neural Networks (RNNs) are used for sequential/temporal data (RNNs learn "when" patterns occur).
Refer to lecture 47, slide 10.

QUESTION 9: [1 mark]

A _______________ is a small matrix that slides over an image to detect patterns like edges, textures, and shapes.

a. kernel
b. stride
c. padding
d. pooling

Correct Answer: a. kernel

Detailed Solution:
A kernel (filter) is a small matrix that slides over an image to detect patterns like edges, textures, and shapes. A stride is how many steps the filter moves at a time. Padding is like adding a border of zeros around the image before applying the filter. Pooling layers down sample feature maps to retain dominant features while reducing computational complexity and overfitting.

Refer to lecture 48, slide 5, 6 and 7.

QUESTION 10: [2 marks]

Match the following activation functions with their descriptions.

Activation function	Description
P. ReLU	i. Output is between 0 and 1 and is good for probabilities
Q. Leaky ReLU	ii. Output is between -1 and 1 and it allows negative outputs
R. Sigmoid	iii. Makes negative values very small and helps avoid dead neurons
S. Tanh	iv. It makes negative values 0 and is best for deep networks.

a. P-iii; Q-iv; R-ii; S-i
b. P-iv; Q-iii; R-i; S-ii
c. P-ii; Q-i; R-iii; S-iv
d. P-i; Q-ii; R-iv; S-iii

Correct Answer: b. P-iv; Q-iii; R-i; S-ii

Detailed Solution: Refer to lecture 48, slide 8.

QUESTION 11: [1 mark]

_______________ is a type of machine learning where an agent learns to make decisions by interacting with an environment.

a. Supervised Learning
b. Unsupervised Learning
c. Reinforcement Learning
d. Neural Networks

Correct Answer: c. Reinforcement Learning

Detailed Solution: Refer to lecture 50, slide 3.

QUESTION 12: [2 marks]

Match the following core components of reinforcement learning with their descriptions.

Core components	Description
P. Agent	i. Feedback from environment
Q. Action	ii. Decision maker
R. Reward	iii. Strategy that maps states to actions
S. Policy	iv. Decision made by agent

a. P-iii; Q-i; R-iv; S-ii
b. P-iv; Q-iii; R-ii; S-i
c. P-i; Q-ii; R-iii; S-iv
d. P-ii; Q-iv; R-i; S-iii

Correct Answer: d. P-ii; Q-iv; R-i; S-iii

Detailed Solution: Refer to lecture 50, slide 4.

QUESTION 13: [2 marks]

Match the types of Reinforcement learning (RL) with their descriptions.

Type of RL	Description
P. Model-Free RL	i. The agent directly learns a policy that maps states to actions, optimizing it to maximize expected rewards.
Q. Policy-Based RL	ii. The agent builds a model of the environment to simulate outcomes before taking actions.
R. Model-Based RL	iii. The agent learns purely from experience without an explicit model of the environment.

a. P-iii; Q-i; R-ii
b. P-ii; Q-iii; R-ii
c. P-i; Q-iii; R-ii
d. P-iii; Q-ii; R-i

Correct Answer: a. P-iii; Q-i; R-ii

Detailed Solution: Refer to lecture 50, slide 9.

QUESTION 14: [2 marks]

_____________ is defined as an expectation over infinite future rewards, and __________ gives a practical recursive way to compute it.

a. Q-value, Boltzmann Equation
b. Q-value, Bellman Equation
c. Bellman Equation, Q-value
d. Boltzmann Equation, Q-value

Correct Answer: b. Q-value, Bellman Equation

Detailed Solution: Refer to lecture 50, slide 7.

********END*******